

How to use PlasmidSurfer

A short, illustrated handbook for infection-control and surveillance teams

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[Team / contact]

START HERE

What PlasmidSurfer does

You have a plasmid from a patient, usually a carbapenem-resistant *K. pneumoniae*, and you want to know three things quickly: is it a known epidemic plasmid, where has its family been, and how worried should we be?

PlasmidSurfer is a map of **10,114 *K. pneumoniae* plasmids** from the public PLSDB database. Each dot is a plasmid. Lines join plasmids that the PLING tool considers **evolutionarily related**, and related plasmids form **communities**, which act as plasmid families. You paste your sequence, PlasmidSurfer finds the closest known plasmids, and shows them on the map with their resistance genes, their family's history and a risk verdict.

Quick start: six steps

1. Open the website and click **PLSDB search** at the top.
2. Paste your plasmid sequence (FASTA), or choose a file. Leave the method on **Mash dist** for a whole plasmid.
3. Click **Run PLSDB search** and wait about a minute. The closest plasmids appear in a table, best first.
4. Tick **Hide hub plasmids**, then click **Show all hits in the graph**. Your hits light up on the map.
5. Click the top hit. The right-hand panel shows its **verdict** (for example High concern), its resistance genes and its family.
6. Under **Community epidemiology**, click **Show on map** and **Show timeline** to see where and when the family has been found, and in which species.

> **No sequence yet?** You can still explore. Type an accession in **Find**, or click any dot in the graph.

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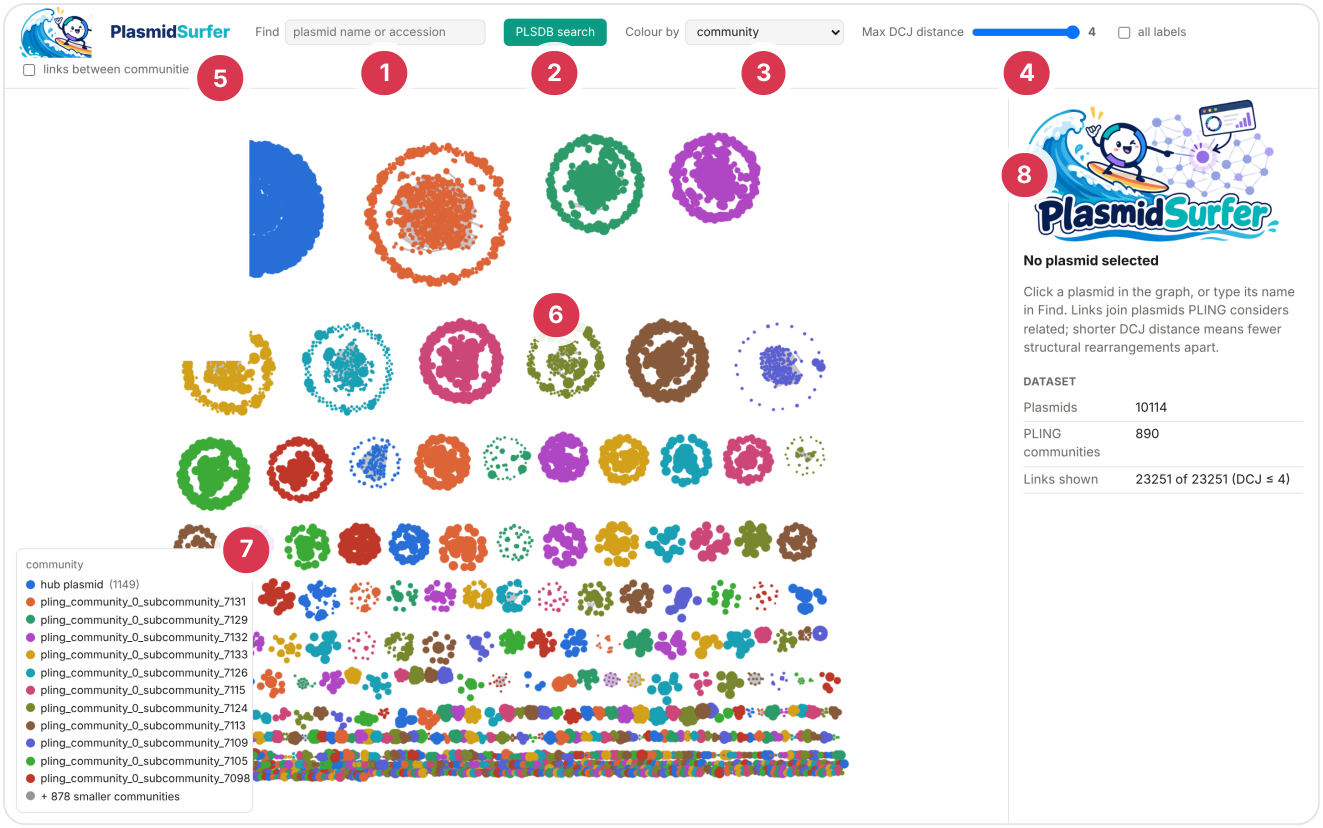
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ORIENTATION

The screen at a glance



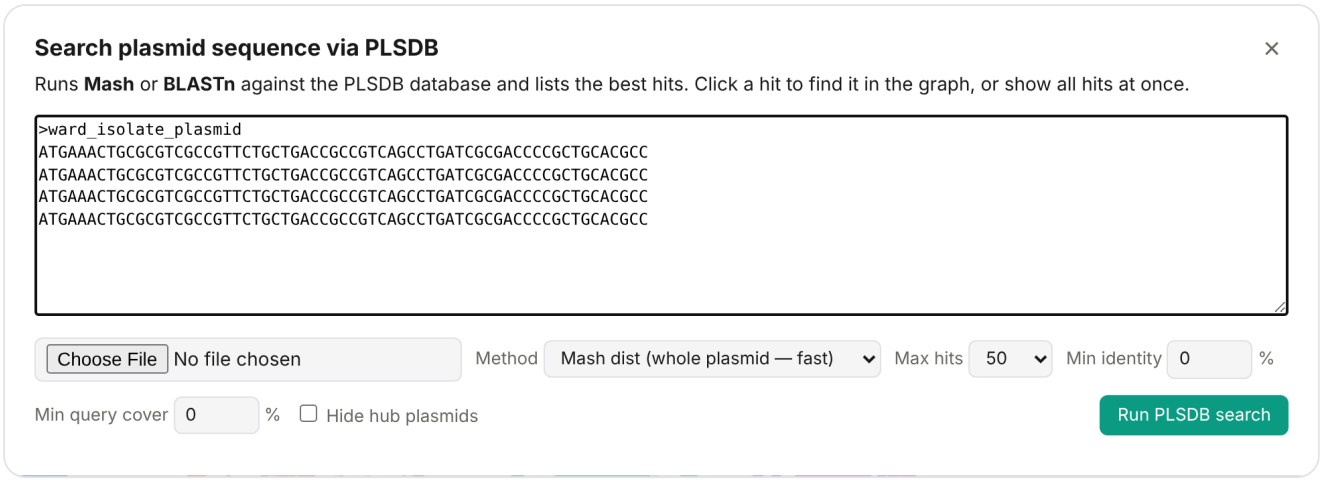
1. **Find** — type a plasmid accession (e.g. `NZ_OQ821115.1`) to jump straight to it.
2. **PLSDB search** — paste your own sequence and find its closest relatives (pages 4–6).
3. **Colour by** — colour the dots by community, risk verdict, AMR tier, mobility, country, year…
4. **Max DCJ distance** — how close two plasmids must be to be linked. Lower = only very close relatives.
5. **Links between communities** — also draw links that cross from one family to another.
6. **The graph** — each blob is a PLING community (a plasmid family); bigger dots are longer plasmids. Scroll to zoom, drag to pan, click a dot to open it.
7. **Legend** — click any entry to select that whole group; click again (or press Esc) to clear.
8. **Side panel** — details of whatever you selected: a plasmid, a group or your search hits.

> The large blue disc on the left is the group of **hub plasmids**: plasmids that PLING links to many families at once. They are real plasmids, but not one family, so treat them separately.

STEP 1

Search a sequence

Click **PLSDB search**. Paste the sequence in FASTA format, or click **Choose file** to load a `.fasta` / `.fa` file. If you paste bare sequence without a `>` header line, one is added for you.



The search window. Your sequence goes in the large box; options sit underneath.

Which method should I use?

METHOD	USE IT WHEN	LIMITS
Mash dist (default)	You have a **whole plasmid** and want the most similar complete plasmids. Fast.	Any length
Mash screen	You want plasmids that **contain** your sequence, e.g. a partial assembly.	Any length
BLASTn	You have a **gene or fragment** (e.g. a *bla*KPC region) and want exact alignments.	100–5,000 bp per record, up to 10 records

Options

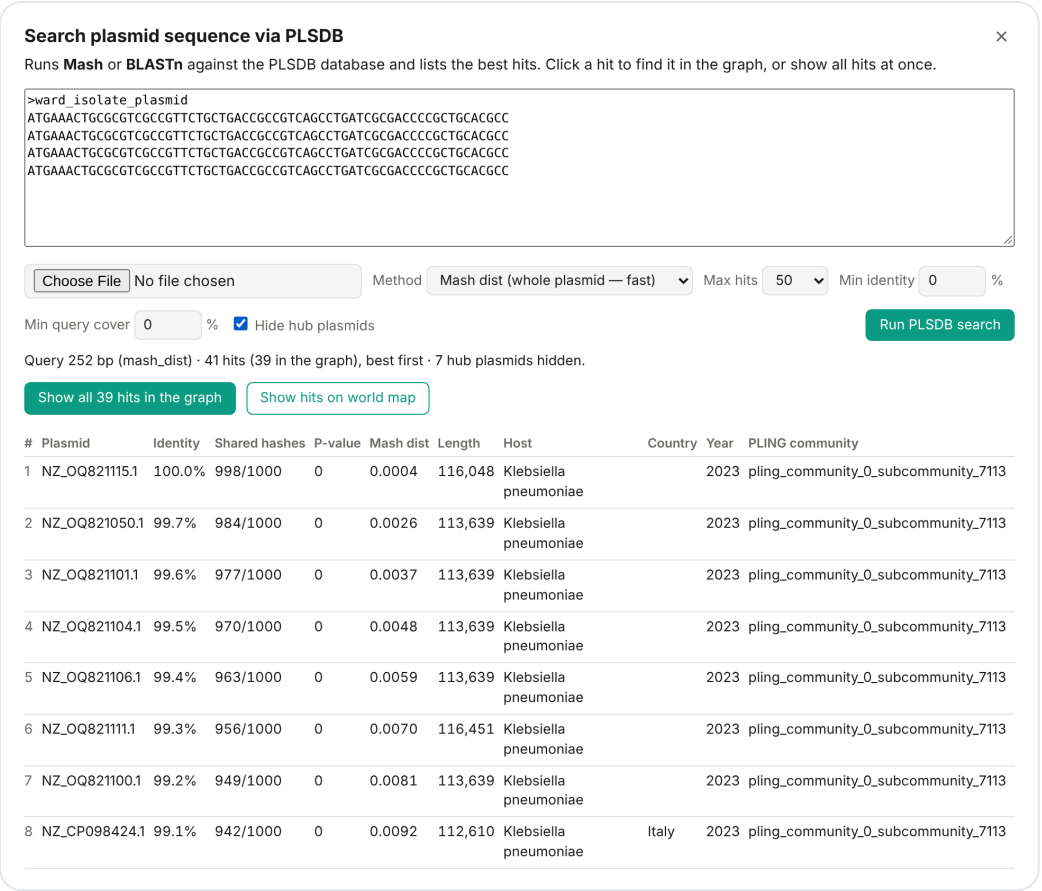
Max hits	How many matches to list (25–250).
Min identity / Min query cover	Drop weak matches. Start at 0; raise identity to ~99 % to keep only near-identical plasmids.
Hide hub plasmids	Removes hub plasmids from the results (recommended, see next page). You can tick it before or after the search.

Click **Run PLSDB search**. The status line shows progress (“Searching PLSDB… 12 s”). Most searches finish within a minute; the page waits up to 10 minutes.

> **Privacy:** your sequence is sent to the public PLSDB service to be searched. Do not paste anything you are not allowed to share outside your institution.

STEP 2

Read the results



Search plasmid sequence via PLSDB

Runs **Mash** or **BLASTn** against the PLSDB database and lists the best hits. Click a hit to find it in the graph, or show all hits at once.

```
>ward_isolate_plasmid
ATGAAACTGCGCGTCGCCGTTCTGCTGACCGCCGTCAGCCTGATCGCGACCCCGCTGCACGCC
ATGAAACTGCGCGTCGCCGTTCTGCTGACCGCCGTCAGCCTGATCGCGACCCCGCTGCACGCC
ATGAAACTGCGCGTCGCCGTTCTGCTGACCGCCGTCAGCCTGATCGCGACCCCGCTGCACGCC
ATGAAACTGCGCGTCGCCGTTCTGCTGACCGCCGTCAGCCTGATCGCGACCCCGCTGCACGCC
```

Choose File No file chosen | Method: Mash dist (whole plasmid — fast) | Max hits: 50 | Min identity: 0 %

Min query cover: 0 % | Hide hub plasmids | Run PLSDB search

Query 252 bp (mash_dist) · 41 hits (39 in the graph), best first · 7 hub plasmids hidden.

Show all 39 hits in the graph | Show hits on world map

#	Plasmid	Identity	Shared hashes	P-value	Mash dist	Length	Host	Country	Year	PLING community
1	NZ_OQ821115.1	100.0%	998/1000	0	0.0004	116,048	Klebsiella pneumoniae		2023	pling_community_0_subcommunity_7113
2	NZ_OQ821050.1	99.7%	984/1000	0	0.0026	113,639	Klebsiella pneumoniae		2023	pling_community_0_subcommunity_7113
3	NZ_OQ821101.1	99.6%	977/1000	0	0.0037	113,639	Klebsiella pneumoniae		2023	pling_community_0_subcommunity_7113
4	NZ_OQ821104.1	99.5%	970/1000	0	0.0048	113,639	Klebsiella pneumoniae		2023	pling_community_0_subcommunity_7113
5	NZ_OQ821106.1	99.4%	963/1000	0	0.0059	113,639	Klebsiella pneumoniae		2023	pling_community_0_subcommunity_7113
6	NZ_OQ821111.1	99.3%	956/1000	0	0.0070	116,451	Klebsiella pneumoniae		2023	pling_community_0_subcommunity_7113
7	NZ_OQ821100.1	99.2%	949/1000	0	0.0081	113,639	Klebsiella pneumoniae		2023	pling_community_0_subcommunity_7113
8	NZ_CP098424.1	99.1%	942/1000	0	0.0092	112,610	Klebsiella pneumoniae	Italy	2023	pling_community_0_subcommunity_7113

Example search (illustrative data). Hits are sorted best first; the status line says how many are in the graph and how many hubs were hidden.

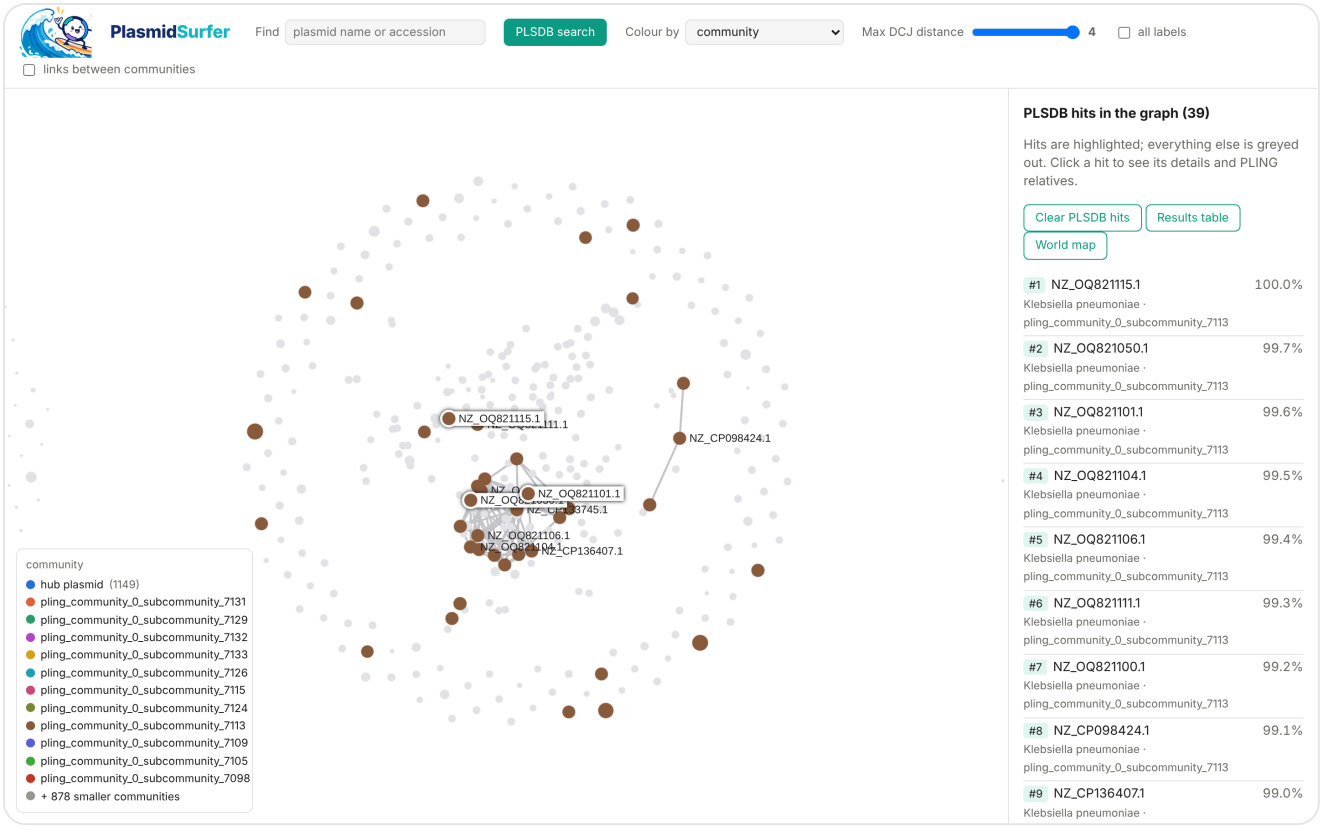
COLUMN	MEANING
Identity	How similar the hit is to your sequence. 99.9 % ≈ the same plasmid.
Shared hashes (Mash) / **Query cover** (BLASTn)	How much of your sequence the match covers. 990/1000 means almost all of it.
Mash dist / E-value	Lower is better.
Host, Country, Year	Where and when that plasmid was recorded.
PLING community	Its plasmid family. *"not in graph"* means PLSDB has it, but it is not one of the 10,114 *K. pneumoniae* plasmids drawn here (often another species).

> **Why hide hub plasmids?** Hubs share pieces with many families, so they often match your query without telling you which family it belongs to. Hiding them lets the real family stand out. Untick the box to bring them back.

Next: click any row to jump to that plasmid, or **Show all hits in the graph** to see them together, or **Show hits on world map** to see their countries.

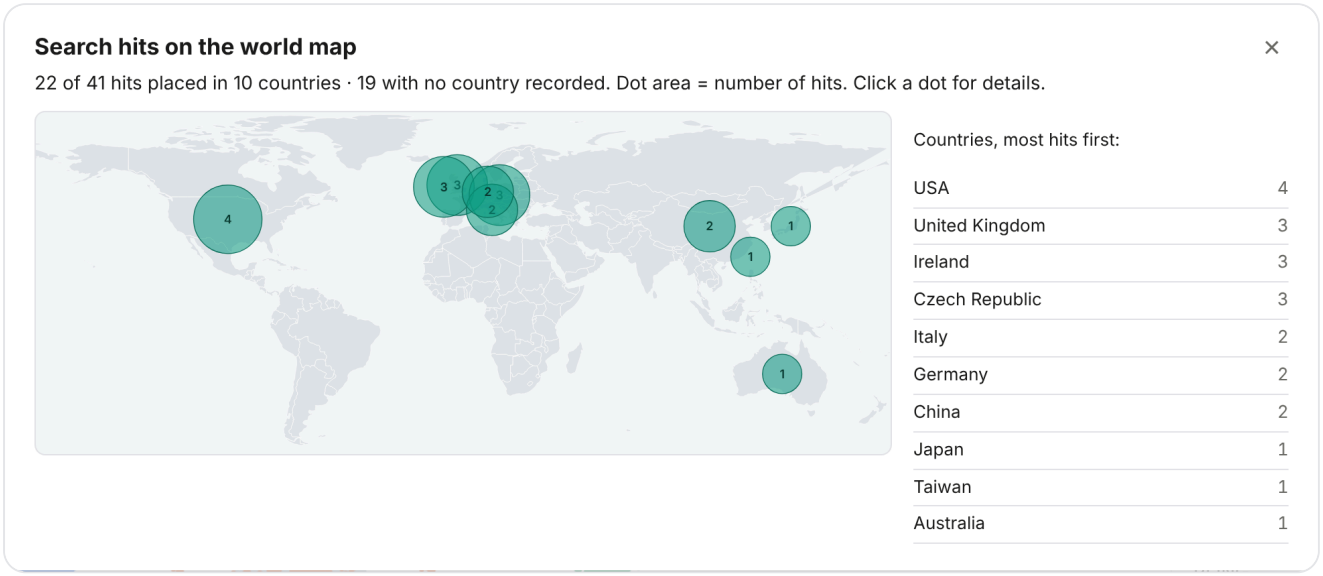
STEP 3

See the hits in the graph and on the map



Hits light up and everything else is greyed out. Here most hits fall in one community: **one plasmid family, not unrelated matches**.

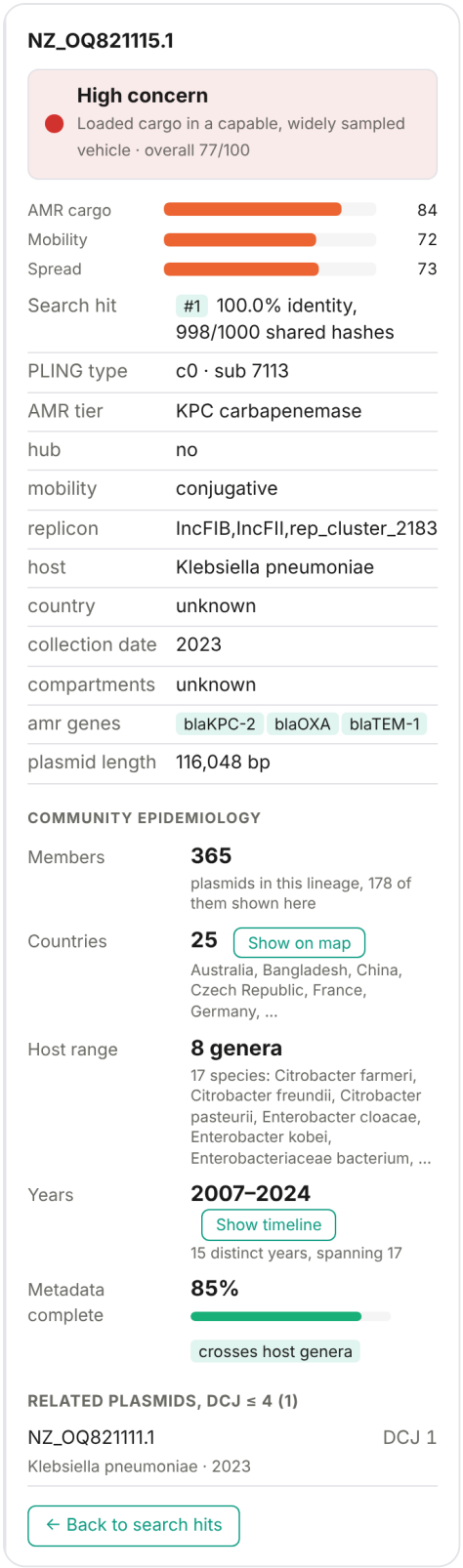
The side panel lists the hits with their rank and identity. Buttons at the top: **Clear PLSDB hits**, **Results table** (reopens the table) and **World map**.



World map of the hits: one dot per country, area = number of hits. Click a dot or a country in the list to see its plasmids, then click one to open it in the graph.

STEP 4

Read a plasmid's panel and verdict



The verdict box

The coloured box gives a one-word answer to *"how worried should we be?"*:

Low concern | Monitor | Concerning | High concern

It combines three scores, shown as bars. Each is a **percentile** among the 10,114 *K. pneumoniae* plasmids: 84 means "more serious than 84 % of them".

AMR cargo	The worst resistance gene it carries (carbapenemases highest).
Mobility	Can it transfer itself (conjugative) or be moved?
Spread	How widely its family has been found.

A red **potential wave** label means: this plasmid is clean today, but its relatives carry carbapenemases and it can move. Watch it.

The details

Search hit (if you came from a search), **PLING type** (its family), **AMR tier** and genes, mobility, replicon, host, country, year, One-Health compartment and length.

Community epidemiology

Facts about the **whole family in PLSDB**, including other species: number of members, countries, host genera and years. **Crosses host genera** means the family has moved between bacterial genera.

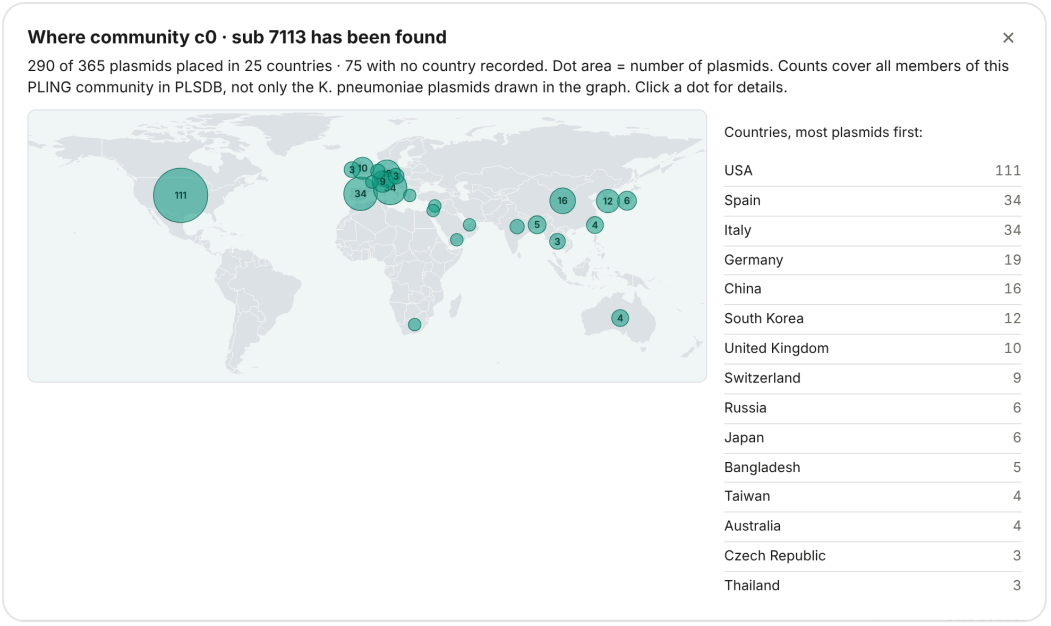
Related plasmids lists its closest relatives with their DCJ distance (1 = one rearrangement apart). Click one to move to it.

STEP 5

Explore the family: map and timeline

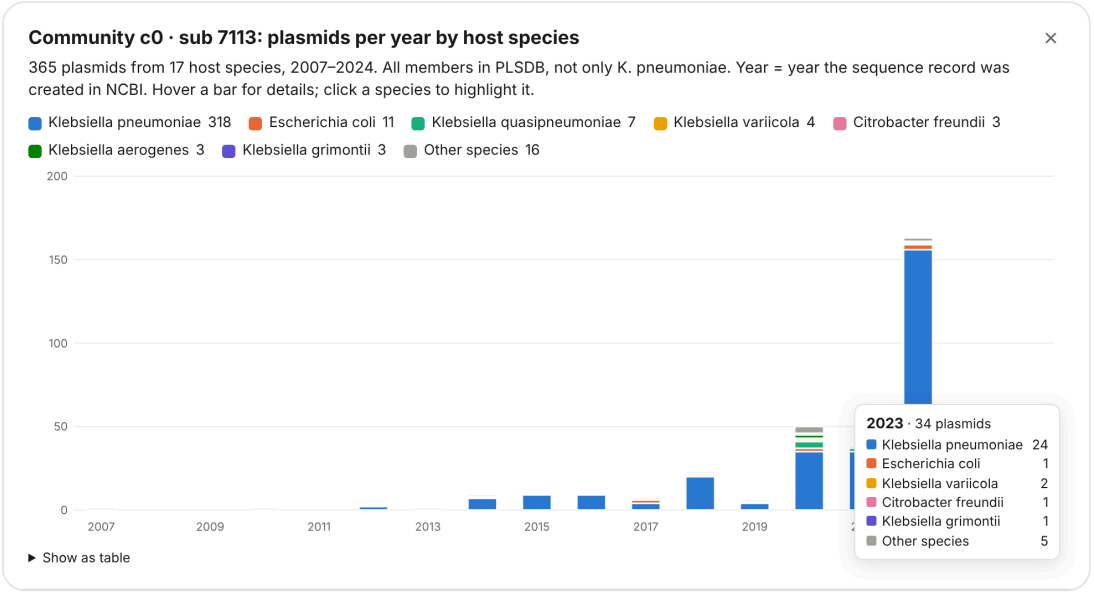
Two buttons in **Community epidemiology** open pop-ups about the plasmid's **whole PLING community** in PLSDB, not only the *K. pneumoniae* members drawn in the graph.

Show on map (next to Countries)



Where the family has been found. Dot area = number of plasmids from that country. Click a country to see how many of its plasmids are in the graph and open them.

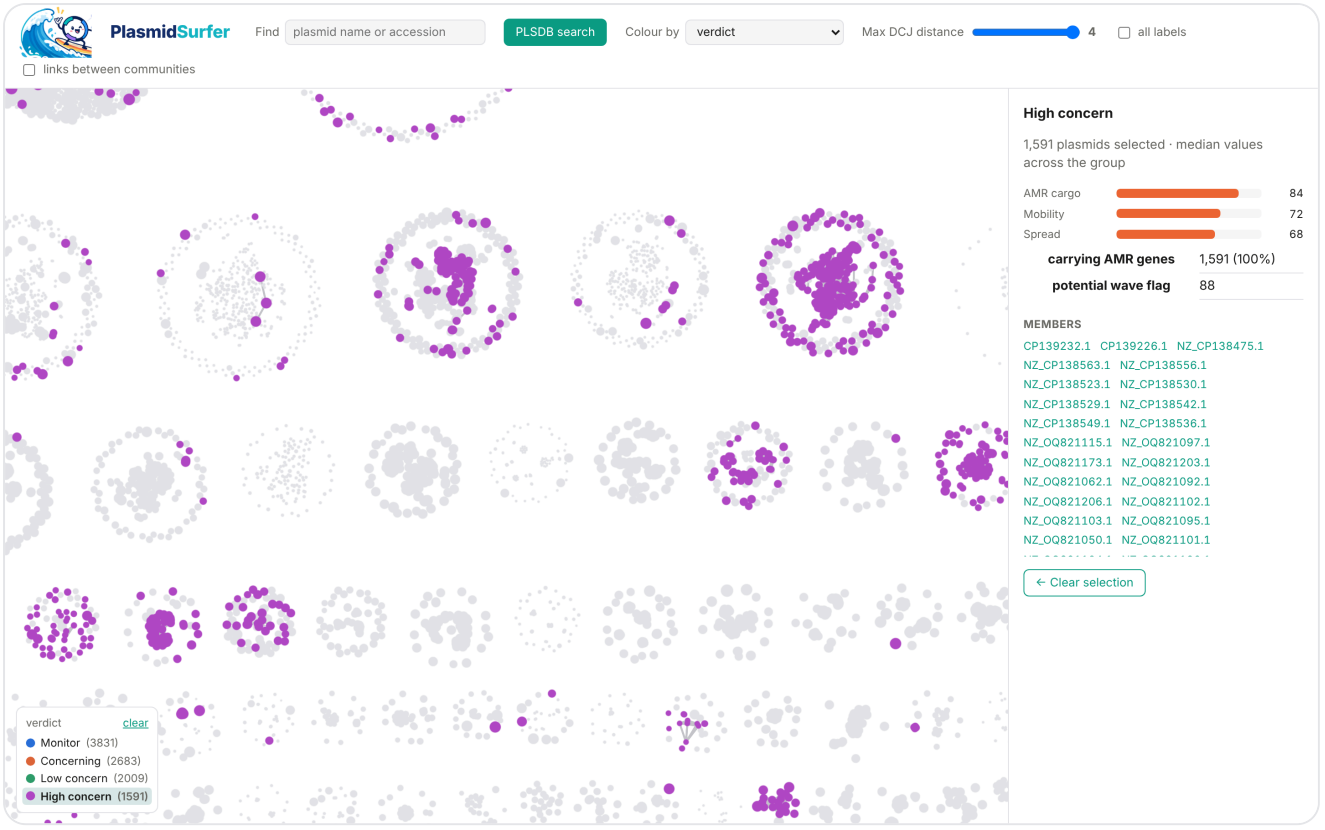
Show timeline (next to Years)



Plasmids per year, coloured by host species (top 7 species; the rest are grey “Other species”). Hover a bar for exact counts; click a species in the legend to highlight it; **Show as table** gives the numbers. A growing bar, or new colours appearing, means the family is expanding or reaching new species.

STEP 6

Explore the whole graph



Colour by **verdict**, then click **High concern** in the legend: all 1,591 high-concern plasmids light up and the panel shows their median scores and a member list.

I WANT TO...	DO THIS
find a known plasmid	Type its accession in **Find** and press Enter.
see high-risk plasmids	**Colour by → verdict**, click **High concern** in the legend.
see all KPC / NDM / OXA-48 plasmids	**Colour by → AMR tier**, click the tier in the legend.
see plasmids from one country or year	**Colour by → country** or **collection year**, click the entry.
see only very close relatives	Lower **Max DCJ distance** to 1 or 2.
see how families connect	Tick **links between communities**.
read every plasmid name	Tick **all labels** (best when zoomed in).
go back	Press **Esc**, click **clear** in the legend, or click an empty part of the graph.

REFERENCE

How the verdict is calculated

The verdict is **rule-based**, with no machine learning, so every number can be traced and challenged. Full details are in `docs/METHODS.md`.

SCORE	BASED ON	WEIGHT
AMR cargo	Worst gene: 2+ carbapenemases > NDM/IMP/VIM > KPC/OXA-48 > *mcr* > ESBL > AmpC > quinolone > other β-lactam > other AMR	45 %
Mobility	MOB-suite: conjugative > mobilisable > indeterminate > non-mobilisable, plus relaxase/MPF/oriT	35 %
Spread	Countries, One-Health compartments and size of its PLING family, plus hub status	20 %

Potential wave adds 15 points. Bands: < 30 Low concern | 30–55 Monitor | 55–75 Concerning | ≥ 75 High concern

Keep in mind

- The verdict is a **genotypic triage aid**, not a clinical prediction. Always combine it with your local epidemiology.
- PLSDB reflects **what has been sequenced**; some countries and settings are over-represented.
- "Year" is when the sequence was **submitted to NCBI**, not when the sample was collected.
- A match in the same family suggests relatedness, not proof of transmission. Confirm with your usual typing.

Troubleshooting

PROBLEM	WHAT TO DO
The graph takes a few seconds to appear	Normal: it loads 10,114 plasmids. Wait ~5–10 s.
"BLASTn record is X bp"	BLASTn takes 100–5,000 bp. Use **Mash dist** for whole plasmids.
Search fails or times out	PLSDB may be busy. Try again later, or try Mash instead of BLASTn.
Hits say "not in graph"	They are in PLSDB but not among the *K. pneumoniae* plasmids drawn here.
Search does nothing when opened from a file	The search needs the online version (it runs through the website's server).

Glossary

PLING — tool that measures plasmid relatedness by counting structural rearrangements (**DCJ distance**). **Community** — a family of closely related plasmids. **Hub plasmid** — a plasmid linked to many families. **Conjugative** — can transfer itself to another cell. **PLSDB** — public database of complete plasmid sequences.